1. Equation of the common tangent touching the circle $(x-3)^2+y^2=9$ and the parabola $y^2=4x$ above X-axis is

NIMCET 2008

(a)$\sqrt{3}y=3x+1$ (b)$\sqrt{3}y=-(x+3)$
(c)$\sqrt{3}y=x+3$ (d)$\sqrt{3}y=-(3x+1)$

2. The angle between the asymptotes of the hyperbola $27x^2-9y^2=24$ is

NIMCET 2010

(a)$60°$ (b)$120°$ (c)$30°$ (d)$150°$

3. If any tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ intercepts equal length l on the axes, then $l=$

NIMCET 2010

(a)a^2+b^2 (b)$\sqrt{(a^2+b^2)}$
(c)$(a^2+b^2)^2$ (d)None of these

4. The vertex of parabola $y^2-8x-x+19=0$ is

NIMCET 2011

(a)(3, 4) (b)(4, 3)
(c)(1, 3) (d)(3, 1)

5. The eccentricity of ellipse $9x^2+5y^2-30y=0$ is

NIMCET 2011

(a)$\frac{1}{3}$ (b)$\frac{2}{3}$ (c)$\frac{3}{4}$ (d)$\frac{1}{4}$

6. The equation of the ellipse with major axis along the X-axis and passing through the points (4, 3) and $(-1,4)$ is

NIMCET 2012

(a)$15x^2+7y^2=247$ (b)$7x^2+15y^2=247$
(c)$16x^2+9y^2=247$ (d)$9x^2+16y^2=247$

7. Focus of the parabola $x^2+y^2-2xy-4(x+y-1)$ is

NIMCET 2012

(a)(1, 1) (b)(1, 2) (c)(2, 1) (d)(0, 2)

8. If e and e' be the eccentricities of a hyperbola and its conjugate, then $\frac{1}{e^2}+\frac{1}{e'^2}$ is equal to

NIMCET 2012

(a)0 (b)1 (c)2 (d)None of these

9. The sum of the focal distances of any point on the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ with eccentricities e, is given by

NIMCET 2013

(a)2ae (b)2b (c)2a (d)2be

10. The locus of intersection of two lines $\sqrt{3}x-y=4k\sqrt{3}$ and $k(\sqrt{3}x+y)=4\sqrt{3}$ for different value of k is a hyperbola. The eccentricity of the hyperbola is

NIMCET 2014

(a)1.5 (b)$\sqrt{3}$ (c)2 (d)$\frac{\sqrt{3}}{2}$

11. If $x=1$ is the directrix of the parabola $y^2=kx-8$, then k is

NIMCET 2014

(a)$\frac{1}{8}$ (b)8 (c)4 (d)$\frac{1}{4}$

12. The condition that the line $lx+my+n=0$ becomes a tangent to the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, is

NIMCET 2014

(a)$a^2l+b^2m+n=0$ (b)$al^2+bm^2=n^2$
(c)$al+bm=n$ (d)$a^2l^2+b^2m^2=n^2$

13. If PQ is a double ordinate of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ such that OPQ is an equilateral triangle, where O is the centre of the hyperbola, then which of the following is true?

NIMCET 2014

(a)$b^2>\frac{-a^2}{\sqrt{3}}$ (b)$b^2>\frac{a^2}{3}$
(c)$b^2<\frac{a^2}{3}$ (d)$b^2<\frac{-a^2}{3}$

14. An equilateral triangle is inscribed in the parabola $y^2=4ax$ such that one of the vertices of the triangle coincides with the vertex of the parabola. The length of the side of the triangle is

NIMCET 2014

(a)$a\sqrt{3}$ (b)$2a\sqrt{3}$ (c)$4a\sqrt{3}$ (d)$8a\sqrt{3}$

15. A normal to the curve $x^2=4y$ passes through the point (1, 2). The distance of the origin from the normal is

NIMCET 2014

(a)$\sqrt{2}$ (b)$2\sqrt{2}$ (c)$\frac{1}{\sqrt{2}}$ (d)$\frac{3}{\sqrt{2}}$

16. The locus of the mid-point of all chords of the parabola $y^2=4x$, which are drawn through its vertex is

NIMCET 2015

(a)$y^2=8x$ (b)$y^2=2x$
(c)$x^2+4y^2=16$ (d)$x^2=2y$

17. The foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ coincide, then value of b^2 is

NIMCET 2015

(a)1 (b)5 (c)7 (d)9

18. If $3x+4y+K=0$ is a tangent to the hyperbola $9x^2-16y^2=144$, then the value of K is NIMCET 2015

(a)0 (b)1 (c)-1 (d)-3

19. The radius of the circle passing through the foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ and having its centre at (0, 3) is

NIMCET 2015

(a)4 units (b)3 units
(c)$\sqrt{12}$ units (d)$\frac{7}{2}$ units

20. The line $3x+5y=k$ touches the ellipse $16x^2+25y^2=400$ if k is

NIMCET 2016

(a)$\pm\sqrt{5}$ (b)$\pm\sqrt{15}$ (c)± 25 (d)$\pm\sqrt{35}$

21. The vertex of the parabola whose focus is $(-1,1)$ and directrix is $4x+3y-24=0$, is

NIMCET 2016

(a)$\left(0,\frac{3}{2}\right)$ (b)$\left(0,\frac{5}{2}\right)$ (c)$\left(1,\frac{3}{2}\right)$ (d)$\left(1,\frac{5}{2}\right)$

22. Two common tangents to the circle $x^2+y^2=2a^2$ and the parabola $y^2=8x$ are

NIMCET 2016

(a)$x=\pm(y+2a)$ (b)$y=\pm(x+2a)$
(c)$x=\pm(y+a)$ (d)$y=\pm(x+a)$

23. The foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ coincide. The value of b^2 is

NIMCET 2016

(a)5 (b)7 (c)9 (d)1

24. The equation of the hyperbola with centre at the origin, length of the transverse axis is 6 and one focus (0, 4) is

NIMCET 2017

(a)$\frac{y^2}{9}+\frac{x^2}{7}=1$ (b)$\frac{y^2}{9}-\frac{x^2}{7}=1$
(c)$\frac{y^2}{7}+\frac{x^2}{9}=1$ (d)$\frac{y^2}{7}-\frac{x^2}{9}=1$

25. Equation of a common tangent with positive slope to the circle $x^2+y^2-8x=0$ as well as to the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$ is

NIMCET 2018

(a)$2x-\sqrt{5}y-20=0$ (b)$2x-\sqrt{5}y+4=0$
(c)$3x-4y+8=0$ (d)$4x-3y+4=0$

26. Equation of the tangent from the point $(3,-1)$ to the ellipse $2x^2+9y^2=3$ is

NIMCET 2019

(a)$2x-3y-3=0$ (b)$2x+3y-3=0$
(c)$2x+y-3=0$ (d)None of these

27. If S and S′ are foci of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$, B is the end of the minor axis and BSS′ is an equilateral triangle, then the eccentricities of the ellipse is

NIMCET 2019

(a)$\frac{1}{2}$ (b)$\frac{1}{3}$ (c)$\frac{1}{4}$ (d)$\frac{1}{5}$

28. The tangent to an ellipse $x^2+16y^2=16$ and making angle 60° with x-axis is

NIMCET 2020

(a)$x-\sqrt{3}y+7=0$ (b)$\sqrt{3}x-y+8=0$
(c)$\sqrt{3}x-y+7=0$ (d)$x+\sqrt{3}y-7=0$

29. Find the number of point(s) of intersection of the ellipse $\frac{x^2}{4}+\frac{(y-1)^2}{9}=1$ and the circle $x^2+y^2=4$.

NIMCET 2020

(a)4 (b)3 (c)2 (d)1

30. If $\left(\frac{x}{a}\right)^2+\left(\frac{y}{b}\right)^2=1$, $(a>b)$ and $x^2-y^2=c^2$ cut at right angles, then

NIMCET 2022

(a)$a^2+b^2=2c^2$ (b)$b^2-a^2=2c^2$
(c)$a^2-b^2=2c^2$ (d)$a^2-b^2=c^2$

31. Coordinate of focus of the parabola $4y^2+12x-20y+67=0$ is

NIMCET 2022

(a)$\left(-\frac{5}{4},\frac{17}{4}\right)$ (b)$\left(-\frac{17}{2},\frac{5}{4}\right)$
(c)$\left(-\frac{17}{4},\frac{5}{2}\right)$ (d)$\left(-\frac{5}{2},\frac{17}{4}\right)$

32. The eccentricities of an ellipse, with its centre at the origin is 1/3. If one of the directrix is $x=9$, then the equation of ellipse is

NIMCET 2022

(a)$9x^2+8y^2=72$ (b)$8x^2+9y^2=72$
(c)$8x^2+7y^2=56$ (d)$7x^2+8y^2=56$

33. If the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{b^2}=1$ are the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ are coincide, then the value of b^2 is

NIMCET 2022

(a)25 (b)16 (c)64 (d)49

34. Find foci of the equation $x^2+2x-4y^2+8y-7=0$

NIMCET 2023

(a)$(\sqrt{5}\pm 1,1)$ (b)$(-1\pm\sqrt{5},1)$
(c)$(-1,\sqrt{5}\pm 1)$ (d)$(1,-1\pm\sqrt{5})$

35. The locus of the mid-point of all chords of the parabola $y^2 = 4x$ which are through its vertex is

NIMCET 2023

(a)$y^2 = 8x$ (b)$y^2 = 2x$
(c)$y^2 + 4y^2 = 16$ (d)$x^2 = 2y$

36. A point P in the first quadrant, lies on $y^2 = 4ax, a > 0$, and keeps a distance of 5a units from its focus. Which of the following points lies on the locus of P?

NIMCET 2023

(a)(1, 0) (b)(1, 1) (c)(0, 2) (d)(2, 0)

37. A circle touches the X -axis and also touches the circle with centre (0 ,3) and radius 2. The locus of the centre of the circle is

(a)a circle (b)An ellipse
(c)a parabola (d)a hyperbola

NIMCET 2015

38. Let a , b , c , d be non zero numbers . If the point of intersection of the line 4ax + 2ay + c =0 and 5bx + 2by + d = 0 lies in the fourth quadrant and is equidistance from the two are then

(a)a+ b + c +d =0 (b)ad –bc =0
(c)3bc -2ad =0 (d)3bc + 2ad =0

NIMCET 2023

39. Area of the parallelogram formed by the lines y= 4x, y = 4x + 1, x +y = 0 and x + y = 1 is

(a)$\frac{1}{5}$ (b)$\frac{2}{5}$ (c)5 (d)10

NIMCET 2022

40. A straight line through the point (4, 5) is such that its intercept between the axes is bisected at A, then its equation is

(a)3x + 4y =20 (b)3x – 4y + 7 = 0
(c)5x -4y = 40 (d)5x + 4y = 40

NIMCET 2022

41. There are two circles in xy-plane whose equations are $x^2 + y^2 - 2y = 0$ and $x^2 + y^2 - 2y - 3 = 0$. A point (x, y)is choosen at random inside the larger circle. then, the probability that the point has been taken from smaller circle is

(a)$\frac{1}{3}$ (b)$\frac{2}{3}$ (c)$\frac{1}{2}$ (d)$\frac{1}{4}$

NIMCET 2022

42. If P(1,2), Q(4,6), R(5,7) and S(a,b) are the vertices of a parallelogram PQRS, then

(a)a=2, b=3 (b)a=3, b=4
(c)a=2, b=4 (d)a=3, b=5

NIMCET 2021

43. The lines $px + qy = 1$ and $qx + py = 1$ are respectively the sides AB, AC of the ΔABC and the base BC is bisected at (p,q). Equation of the median of the triangle through the vertex A is

(a)(2pq-1)(qx+py-1)-(p^2+q^2-1)(px+qy-1)=0
(b)(2pq-1)(px+qy-1)+(p^2+q^2-1)(qx+py-1)=0
(c)(2pq-1)(px+qy-1)-(p^2+q^2-1)(qx+py-1)=0
(d)(2pq-1)(qx+qy-1)+(p^2+q^2-1)(px+qy-1)=0

NIMCET 2021

44. In a triangle, if the sum of two sides is x and their product is y such that (x+ z)(x-z)=y, where z is the third side of the triangle, then triangle is

(a)equilateral (b)right angled
(c)isosceles (d)obtuse angled

NIMCET 2021

45. The number of common tangents to the circle $x^2 + y^2 = 4$ and $x^2 + y^2 - 6x - 8y = 24$ is

(a)0 (b)1 (c)3 (d)4

NIMCET 2021

46. If two circle $x^2 + y^2 + 2gx + 2fy = 0$ and $x^2 + y^2 + 2g'x + 2f'y = 0$ touch each other, then which of the following is true?

(a)$gf = gf'$ (b)$g'f = gf'$
(c)$gg' = ff'$ (d)None of these

NIMCET 2015

47. The foot of the perpendicular from the point(2,4) upon $x + y = 1$ is

(a)$\left(\frac{1}{2}, \frac{3}{2}\right)$ (b)$\left(-\frac{1}{2}, \frac{3}{2}\right)$ (c)$\left(\frac{4}{3}, \frac{1}{2}\right)$ (d)$\left(\frac{4}{3}, -\frac{1}{2}\right)$

NIMCET 2015

48. If $(2,1), (-1,-2), (3,3)$ are the mid-points of the sides BC, CA and AB of a triangle ABC, then equation of the line BC is

(a)$5x + 4y + 6 = 0$ (b)$5x - 4y - 6 = 0$
(c)$5x + 4y - 6 = 0$ (d)$5x - 4y + 6 = 0$

NIMCET 2015

49. If $(-4,5)$ is one vertex and $7x - y + 8 = 0$ is one diagonal of a square, then the equation of the other diagonal is

(a)$x + 7y = 21$ (b)$x + 7y = 31$
(c)$x + 7y = 31$ (d)$x + 7y = 35$

NIMCET 2015

50. A circle touches the X- axis and also touches another circle with centre of the first circle is
(a)a parabola (b)a hyperbola
(c)a circle (d)an ellipse

NIMCET 2015

51. For two circles $x^2 + y^2 = 16$ and $x^2 + y^2 - 2y = 0$, there is/are
(a)One pair of common tangent
(b)Two pair of common tangents
(c)Three pair of common tangents
(d)No common tangents

NIMCET 2019

52. The equation of circle passing through the point (4,6) and whose diameters are along $x + 2y - 5 = 0$ and $3x - y - 1 = 0$ is
(a)$x^2 + y^2 - 2x - 6y - 20 = 0$
(b) $x^2 + y^2 - 6x - 2y - 20 = 0$
(c) $x^2 + y^2 - 2x - 4y - 20 = 0$
(d) $x^2 + y^2 - 4x - 2y - 20 = 0$

NIMCET 2019

53. If the lines $x + (a-1)y + 1 = 0$ *and* $2x + a^2y - 1 = 0$ are perpendicular, then the condition satisfied by a is **NIMCET 2017**
(a)$|a| = 2$ (b)$0 < a < 1$
(c)$-1 < a < 0$ (d) $a = -1$

54. If $x^2 + 3xy + 2y^2 - x - 4y - 6 = 0$ represents a pair of straight lines, their point of intersection is
(a)(0,0) (b)(8,5) (c) (8,-5) (d)(-2 5)

NIMCET 2017

55. The two parabolas $y^2 = 4a(x + c)$ and $y^2 = 4bx$, a > b > 0 cannot have a common normal unless (Parabola)
(a) c > 2 (a + b) (b) c > 2(a – b)
(c) c < 2(a – b) (d) c < 2/ a-c

NIMCET 2024

56. At how many points the following curves intersect (Ellipse) **NIMCET 2024**
$\frac{y^2}{9} - \frac{x^2}{16} = 1$ and $\frac{x^2}{4} + \frac{(y-4)^2}{16} = 1$
(a) 0 (b) 1
(c) 2 (d) 4

57.If (4, 3) and (12, 5) are the two focii of an ellipse passing through the origin, then the eccentricity of the ellipse is (Ellipse)
(a) $\frac{\sqrt{13}}{9}$ (b) $\frac{\sqrt{13}}{18}$
(c) $\frac{\sqrt{17}}{18}$ (d) $\frac{\sqrt{17}}{9}$

NIMCET 2024

58. The equation $3x^2 + 10xy + 11y^2 + 14x + 12y + 5 = 0$ represent, (Ellipse) **NIMCET 2024**
(a) a circle (b) an ellipse
(c) a hyperbola (d) a parabola

59. The circle $x^2 + y^2 + \alpha x + \beta y + \gamma = 0$ is the image of the circle $x^2 + y^2 - 6x - 10y + 30 = 0$ across the line $3x + y = 2$. The value of $\lfloor \alpha + \beta + \gamma \rfloor$ is (where $\lfloor . \rfloor$ represents the floor function) **NIMCET 2025**
(a) 23 (b) 22
(c) 20 (d) 21

60. A circle with its center in the first quadrant touches both the coordinate axes and the line x – y – 2 = 0. Then the area of the circle is **NIMCET 2025**
(a) 2π (b) π (c) $\frac{\pi}{2}$ (d) 4π

61. An equilateral triangle is inscribed in the parabola $y^2 = x$. One vertex of the triangle is at the vertex of the parabola. The centroid of triangle is **NIMCET 2025**
(a) $(2, 0)$ (b) $(\sqrt{2}, 0)$
(c) $(1, 0)$ (d) $(\sqrt{3}, 0)$

62. Let F_1, F_2 be the fact of the hyperbola
$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, a > 0, b > 0,
and let O be the origin. Let M be an arbitrary point on curve C and above X-axis and H be a point on MF_1 such that
$MF_2 \perp F_1F_2$, $MF_1 \perp OH$, $|OH| = \lambda |OF_2|$ with λ ϵ (2/5, 3/5), then the range of the eccentricity e in **NIMCET 2025**
(a) $(1, \sqrt{7/3})$ (b) $(\sqrt{7/3}, 2)$
(c) $(\sqrt{2}, \sqrt{3})$ (d) $(\sqrt{3}, 2)$

63. Let the line $\frac{x}{4} + \frac{y}{2} = 1$ meets the x-axis and y-axis at A and B, respectively. M is the midpoint of side AB, and M' is the image of the point M across the line x + y = 1. Let the point P lie on the line x + y = 1 such that the ΔABP is an isosceles triangle with AP = BP. Then the distance between M' and P is **NIMCET 2025**
(a) $\frac{\sqrt{7}}{3}$ (b) $\frac{2\sqrt{5}}{3}$
(c) $\frac{2\sqrt{7}}{3}$ (d) $\frac{\sqrt{5}}{3}$

ANSWER KEY

1.	c	2.	b	3.	b	4.	a	5.	b
6.	b	7.	a	8.	b	9.	c	10.	c
11.	c	12	d	13.	b	14.	d	15.	d
16.	b	17.	c	18.	a	19.	a	20.	c
21.	d	22.	b	23.	b	24.	b	25.	b
26.	b	27.	a	28.	c	29.	b	30.	c
31.	c	32.	b	33.	b	34.	b	35.	b
36.	b	37.	c	38.	c	39.	a	40.	d
41.	d	42.	a	43.	c	44.	d	45.	b
46.	c	47.	b	48.	b	49.	b	50.	a
51.	d	52.	c	53.	d	54.	c	55.	b
56.	c	57.	d	58.	b	59.	a	60.	a
61.	a	62.	b	63.	b				

Analysis 2007 to 2025
Questions Asked in NIMCET 2007 to 2025
PARABOLA

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	00	NIMCET 2016	02
NIMCET 2008	01	NIMCET 2017	00
NIMCET 2009	00	NIMCET 2018	00
NIMCET 2010	00	NIMCET 2019	00
NIMCET 2011	01	NIMCET 2020	00
NIMCET 2012	01	NIMCET 2021	00
NIMCET 2013	00	NIMCET 2022	01
NIMCET 2014	01	NIMCET 2023	02
NIMCET 2015	01	NIMCET 2024	01
		NIMCET 2025	01

ELLIPSE

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	00	NIMCET 2016	02
NIMCET 2008	00	NIMCET 2017	00
NIMCET 2009	00	NIMCET 2018	00
NIMCET 2010	01	NIMCET 2019	02
NIMCET 2011	01	NIMCET 2020	02
NIMCET 2012	01	NIMCET 2021	00
NIMCET 2013	01	NIMCET 2022	03
NIMCET 2014	01	NIMCET 2023	00
NIMCET 2015	01	NIMCET 2024	03
		NIMCET 2025	03

HYPERBOLA

Years	No. of Questions	Years	No. of Questions
NIMCET 2007	00	NIMCET 2016	01
NIMCET 2008	00	NIMCET 2017	01
NIMCET 2009	00	NIMCET 2018	01
NIMCET 2010	01	NIMCET 2019	00
NIMCET 2011	00	NIMCET 2020	00
NIMCET 2012	01	NIMCET 2021	00
NIMCET 2013	00	NIMCET 2022	02
NIMCET 2014	02	NIMCET 2023	01
NIMCET 2015	02	NIMCET 2024	01
		NIMCET 2025	01

SOLUTION

1. **(c)** Parabola $y^2 = 4x$ Circle $(x-3)^2 + y^2 = 9$

$\therefore a = 1$ Center (3,0), Radius: $r = 3$

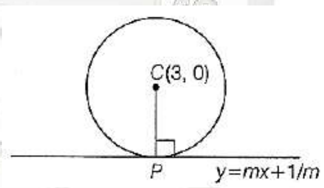
Let tangent of parabola be : $y = mx + \frac{a}{m}(m = \text{slope})$

$y = mx + \frac{1}{m} (\because a = 1)$

Also Eq. (i) is a tangent to the given circle

$\therefore CP = r = 3$

$\left|\frac{3m - 0 + \frac{1}{m}}{\sqrt{m^2+1}}\right| = 3$



$\left(3m + \frac{1}{m}\right)^2 = 9(1 + m^2)$

$\Rightarrow 9m^2 + \frac{1}{m^2} + 6 = 9 + 9m^2$

$\Rightarrow m^2 = \frac{1}{3} \Rightarrow m = \pm\frac{1}{\sqrt{3}} \Rightarrow \sqrt{3}y = \pm(x+3)$

Now, we need a common tangent above X-axis.

$\therefore y > 0$

$\therefore \sqrt{3}y = (x+3)$

2. **(b)** $\frac{2\pi}{3} = 120^0$

3. **(b)** Let equation of tangent to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be

$y = mx \pm \sqrt{a^2m^2 + b^2}$

Also, this tangent equally intercepts the areas

$\therefore m = \pm 1$

$\therefore y = 1\left(x + \sqrt{a^2 + b^2}\right)$

Now, for length of intercept ' l '

Put $x = 0 : y = \sqrt{a^2 + b^2}$ $B = \left(0, \sqrt{a^2 + b^2}\right)$

Now put $y = 0$,

$x = -\sqrt{a^2 + b^2}$ $A = \left(-\sqrt{a^2 + b^2}, 0\right)$

Hence, intercept of equal length l on the axes is

$= \sqrt{a^2 + b^2}$

4. **(a)** $y^2 - 8y - x + 19 = 0$

$(y-4)^2 = x - 19 + 16$

$(y-4)^2 = (x-3)$.

on comparing with parabola $y^2 = 4ax$

then, vertex of parabola $(y-4)^2 = x - 3$

$x_1 = 3$, $y_1 = 4$

vertex is (3, 4)

5. **(b)** $9x^2 + 5y^2 - 30y = 0$

$\Rightarrow 9x^2 + 5(y^2 - 6y) = 0$

$\Rightarrow 9x^2 + 5(y^2 - 6y + 9) = 45$

$\Rightarrow \frac{x^2}{5} + \frac{(y-3)^2}{9} = 1$

$\therefore a^2 = 5, b^2 = 9 (b > a)$

$e^2 = 1 - \frac{a^2}{b^2}$

$e^2 = 1 - \frac{5}{9}$

$\Rightarrow e = \frac{2}{3}$

6 **(b)** Let equation of ellipse be $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

($\because$ major axis is along X-axis)

Now, $(4,3)$ and $(-1,4)$ lies on the curve.

$\therefore S_{(4,3)}: \frac{16}{a^2} + \frac{9}{b^2} = 1$ .... (i)

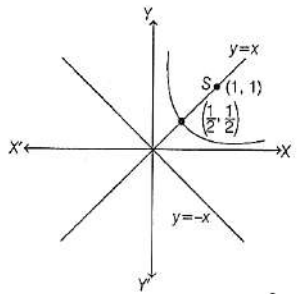
$S_{(-1,4)}: \frac{1}{a^2} + \frac{16}{b^2} = 1$ .... (ii)

Eq. (i) $-16 \times$ Eq. (ii)

$b^2 = \frac{247}{15}$ and $a^2 = \frac{247}{7}$

$\therefore$ Required equation $7x^2 + 15y^2 = 247$

7. **(a)** $x^2 + y^2 - 2xy - 4(x+y-1) = 0$



$\Rightarrow (x-y)^2 = 4\{(x+y) - 1\} \Rightarrow \frac{(x-y)^2}{2} = \frac{4(x+y-1)}{2}$

$\left(\frac{(x-y)}{\sqrt{2}}\right)^2 = \frac{2\sqrt{2}(x+y-1)}{\sqrt{2}}$

Now, we can assume it as $Y^2 = 2\sqrt{2}X$

For finding the vertex $X = 0$ and $Y = 0$

Here,

$x - y = 0$ .... (i)

And

$x + y = 1$ ....(ii)

On solving, we get $x = y = \frac{1}{2}$

$\therefore$ Vertex of parabola $= \left(\frac{1}{2}, \frac{1}{2}\right)$

The focus is the point of intersection of $X = a$ and $Y = 0$

$a = \frac{2\sqrt{2}}{4} = \frac{1}{\sqrt{2}}$

Hence, $\frac{x+y-1}{\sqrt{2}} = \frac{1}{\sqrt{2}}$ and $x - y = 0$

Solving these two equations we will get focus (1,1).

8. **(b)** $e \rightarrow \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ Hyperbola

$e' \rightarrow \frac{-x^2}{a^2} + \frac{y^2}{b^2} = 1$ Conjugate hyperbola

$e = \sqrt{1 + \frac{b^2}{a^2}}, \quad e' = \sqrt{1 + \frac{a^2}{b^2}}$

$\frac{1}{e^2} + \frac{1}{e'^2} = \frac{1}{1+\frac{b^2}{a^2}} + \frac{1}{1+\frac{a^2}{b^2}} = \frac{a^2}{a^2+b^2} + \frac{b^2}{a^2+b^2}$

$= \frac{a^2+b^2}{a^2+b^2} = 1 \qquad \frac{1}{e^2} + \frac{1}{e'^2} = 1$

9. **(c)** Focal property PF' + PF = 2a

10. **(c)**

$\sqrt{3}x - y = 4k\sqrt{3}$

$k(\sqrt{3}x + y) = 4\sqrt{3}$

From Eq (i) $k = \frac{\sqrt{3}x - y}{4\sqrt{3}}$

From Eq (ii) $k = \frac{4\sqrt{3}}{\sqrt{3}x + y}$

$\frac{4\sqrt{3}}{\sqrt{3}x+y} = \frac{\sqrt{3}x-y}{4\sqrt{3}}$

$3x^2 - y^2 = 48, b > a$

$\frac{x^2}{16} - \frac{y^2}{48} = 1$

$\therefore a^2 = 16, b^2 = 48$

$E = \sqrt{1 + \frac{48}{16}} = 2$

11. **(c)** $y^2 = kx - 8$

$y^2 = k\left(x - \frac{8}{k}\right) = 4\left(\frac{k}{4}\right)\left(x - \frac{8}{k}\right)$

$y^2 = 4AX \Rightarrow A = \frac{k}{4}$

Directrix : $x = -A + \frac{8}{k} = 1$ (Given)

$\frac{-k}{4} + \frac{8}{k} = 1$

$\Rightarrow \frac{-k^2+32}{4k} = 1$

$k^2 + 4k - 32 = 0$

$\Rightarrow (k+8)(k-4) = 0$

$k = -8, k = 4$

12. **(d)** Equation of tangent to ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$lx + my + n = 0$

We know the equation of tangent to ellipse is

$bx\cos\theta + ay\sin\theta - ab = 0$

Comparing these two equations

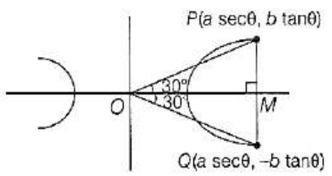
$\frac{b\cos\theta}{l} = \frac{a\sin\theta}{m} = \frac{-ab}{n}$

Hence, $\cos\theta = -\frac{al}{n}, \sin\theta = -\frac{bm}{n}$

$\cos^2\theta + \sin^2\theta = 1$

Hence, $a^2l^2 + b^2m^2 = n^2$

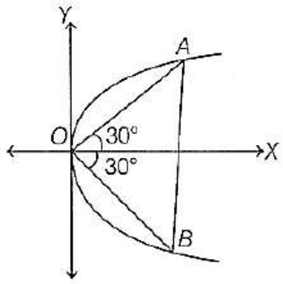
13. **(b)** $\tan 30° = \frac{b\tan\theta}{a\sec\theta}$



$\frac{1}{\sqrt{3}} = \frac{b}{a}\sin\theta \Rightarrow \sin\theta = \frac{a}{b\sqrt{3}} \Rightarrow \sin^2\theta = \frac{a^2}{3b^2}$

$0 < \sin^2\theta < 1 \Rightarrow \frac{a^2}{3b^2} < 1 \Rightarrow \frac{a^2}{3} < b^2$

14. (d) Slope of $OA = \tan 30° = \frac{1}{\sqrt{3}}$



$\Rightarrow$ Equation of $OA: y = \frac{1}{\sqrt{3}}x$

Putting in parabola

$\frac{x^2}{3} = 4ax \Rightarrow x = 12a \Rightarrow y = 4\sqrt{3}a$

$\Rightarrow AB = 2 \times 4\sqrt{3}a = 8\sqrt{3}a$

15. (d) A normal on the curve $x^2 = 4y$ passes through (1,2) and its distance from origin.

Any point on the curve $x^2 = 4by$ is $(2bt, bt^2)$

$\therefore$ Point on $x^2 = 4y$

x = 2t y = t²

$\frac{dx}{dt} = 2$

$\frac{dy}{dx} = \frac{2t}{2} = tr$ $= \frac{dy}{dt} = 2t$

Slope of Normal $= \frac{-1}{t^2}$

$y - t^2 = \frac{-1}{t}(x - 2t) \Rightarrow yt - t^3 = -x + 2t$

$x + yt = t^3 + 2t$

It passes through $(1,2) 1 + 2t = t^3 + 2t \Rightarrow t = 1$

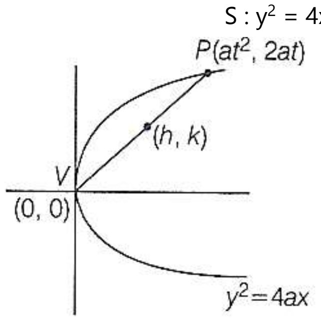
Now normal at $t = 1$

$x + y = 1 + 2 \Rightarrow x + y = 3$

Distance from origin (0,0) is $\frac{|1\times0+1\times0-3|}{\sqrt{1+1}} = \frac{3}{\sqrt{2}}$

16. (b) Let (h, k) be the mid point of chord

S : y² = 4x



$h = \frac{(at^2+0)}{2}, k = \frac{2at+0}{2}$

Hence, $t = \frac{k}{a}$

Put this value in $h = \frac{(at^2)}{2}$

$\Rightarrow 2h = \frac{ak^2}{a^2}$

$\Rightarrow 2ah = k^2$

Put $a = 1$, we get $k^2 = 2h$

Hence, locus $y^2 = 2x$

17. (c)

$\frac{x^2}{16} + \frac{y^2}{b^2} = 1$ (Ellipse); $\frac{x^2}{144} - \frac{y^2}{81} = \frac{1}{25}$ (Hyperbola)

a= 4, A$= \frac{12}{5}, B = \frac{9}{5}$

$e^2 = 1 - \frac{b^2}{a^2}$ $e^2 = 1 + \frac{B^2}{A^2}$

$16 - 16e^2 = b^2$

$e^2 = 1 + \frac{81}{144} \Rightarrow \frac{15}{12} = e$

Focus of ellipse : $(ae, 0)$ Focus of hyperbola : $(Ae, 0)$

$(4e, 0)$ $\left(\frac{12}{5} \times \frac{15}{12}, 0\right) = (3,0)$

Given : $(4e, 0) = (3,0)$

$4e = 3 \Rightarrow e = \frac{3}{4}$

$\therefore b^2 = 16 - 16\left(\frac{9}{16}\right)$

$b^2 = 7$

18. (a) $3x + 4y + K = 0$ is a tangent to the hyperbola

$9x^2 - 16y^2 = 144$

$\frac{x^2}{16} - \frac{y^2}{9} = 1$

$a = 4, b = 3$

Given tangent $y = \frac{-3}{4}x - \frac{K}{4}$

Let the line be, $y = mx + c$

$\therefore m = \frac{-3}{4}, c = \frac{-K}{4}$

Condition of Tangency: $c^2 = a^2m^2 - b^2$

$\frac{K^2}{16} = 16 \times \frac{9}{16} - 9$

$\therefore$ K = 0

19. (a) Ellipse : $\frac{x^2}{16} + \frac{y^2}{9} = 1$

$a^2 = 16, b^2 = 9 \Rightarrow b^2 = a^2(1 - e^2) \Rightarrow 9 = 16(1 - e^2)$

$e^2 = 1 - \frac{9}{16} \Rightarrow e = \frac{\sqrt{7}}{4}$

Foci $(\pm\sqrt{7}, 0)$

$\therefore$ Circle passes through $(\sqrt{7}, 0)$& having centre (0,3)

$\therefore r = \sqrt{(\sqrt{7})^2 + (3)^2} = \sqrt{7+9} = 4$ units

20. (c) $3x + 5y = k$ touches the ellipse $16x^2 + 25y^2 = 400$

$\frac{x^2}{25} + \frac{y^2}{16} = 1$

$a = 5, b = 4$

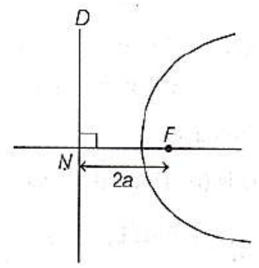
Given line: $y = \frac{-3}{5}x + \frac{k}{5} \Rightarrow m = \frac{-3}{5}, c = \frac{k}{5}$

Condition of tangency: $c^2 = a^2m^2 + b^2$

$\frac{k^2}{25} = 25 \times \frac{9}{25} + 16$

$\Rightarrow k^2 = 625 \Rightarrow k = \pm 25$

21. (d) $F(-1,1)$, directrix : $4x + 3y - 24 = 0$



FN = 2a

$\left|\frac{4(-1)+3(1)-24}{\sqrt{4^2+3^2}}\right| = 2a$

$\left|\frac{-4+3-24}{5}\right| = 2a$

$\Rightarrow\ a = \frac{5}{2}$

Slope of directrix $= \frac{-4}{3}$

Slope of $FN = \frac{3}{4}$

Hence equation of $FN, y-1 = \frac{3}{4}(x+1)$

Solving these two equations, we get N

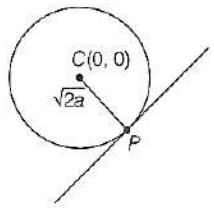
$x = 3, y = 4$

$\therefore N(3,4)$

Now, vertex is mid-point of F and N.

$\therefore\ V(x,y) = \left(\frac{3+(-1)}{2}, \frac{4+1}{2}\right) \Rightarrow V = \left(1, \frac{5}{2}\right)$

22. (b) $x^2+y^2 = 2a^2; P = y^2 = 8ax,$



Standard parabola

$Y^2 = 4AX \Rightarrow A = 2a$

Let equation of tangent to the parabola be: $y = mx + \frac{2a}{m}$

Now, this is also a tangent to the given circle

$\therefore CP = r = \sqrt{2}a$

$= \left|\frac{\frac{2a}{m}}{\sqrt{1+m^2}}\right| = \sqrt{2}a \Rightarrow \frac{4}{m^2(1+m^2)} = 2$

$\Rightarrow m^4+m^2-2 = 0$

$\Rightarrow\ (m^2+2)(m^2-1) = 0 \Rightarrow m = \pm 1$

$\therefore\ y = \pm(x+2a)$

23 (b) Refer to solution 17.

24 (b) Since, focus is (0,4) it will be a conjugate hyperbola.

Let equation of hyperbola be $-\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$

Transverse axis $2b = 6, b = 3$

Focus $(0, be) = (0,4)$

$be = 4 \Rightarrow e = \frac{4}{3}$

$\therefore\ e^2 = 1+\frac{a^2}{b^2}$

$\left(\frac{16}{9}-1\right)\times 9 = a^2 \Rightarrow a^2 = 7$

$\therefore$ Required equation: $\frac{y^2}{9}-\frac{x^2}{7} = 1$

25. (b) Hyperbola $\frac{x^2}{9}-\frac{y^2}{4} = 1$

Equation of its tangent $(bx\sec\theta - ay\tan\theta = ab)$

Here $b = 2, a = 3$

Hence equation of tangent

$2x\sec\theta - 3y\tan\theta = 6$

Equation of circle $x^2+y^2-8x = 0$, center $(4,0)$, radius $= 4$

For common tangent prependicular from center of circle to given line it should be equal to radius.

$\left|\frac{8\sec\theta-6}{\sqrt{4\sec^2\theta+9\tan^2\theta}}\right| = 4$

$\Rightarrow (4\sec\theta-3)^2 = 4(4\sec^2\theta+9\tan^2\theta)$

$\Rightarrow 12\sec^2\theta+8\sec\theta-15 = 0$

$\Rightarrow \sec\theta = \left(-\frac{3}{2}, \frac{5}{6}\right)$ But $\sec\theta \neq \frac{5}{6}$

If $\sec\theta = -\frac{3}{2}$ then $\tan\theta = \frac{\sqrt{5}}{2}$

Put these values in equation of tangent, we get

$2x-\sqrt{5}y+4 = 0$

26. (b) $2x^2+9y^2 = 3$

$\frac{x^2}{\frac{3}{2}}+\frac{y^2}{\frac{1}{3}} = 1, a^2 = \frac{3}{2}, b^2 = \frac{1}{3}$

Let equation of tangent be: $y = mx \pm \sqrt{a^2m^2+b^2}$

$y = mx \pm \sqrt{\frac{3}{2}m^2+\frac{1}{3}}$

Given: $(3,-1)$ is a point on this tangent

$\therefore\ \ -1 = 3m \pm \sqrt{\frac{3}{2}m^2+\frac{1}{3}}$

$3m+1 = \pm\sqrt{\frac{3}{2}m^2+\frac{1}{3}} \Rightarrow (3m+1)^2 = \left(\frac{3}{2}m^2+\frac{1}{3}\right)$

$9m^2+1+6m = \left(\frac{3}{2}m^2+\frac{1}{3}\right)$

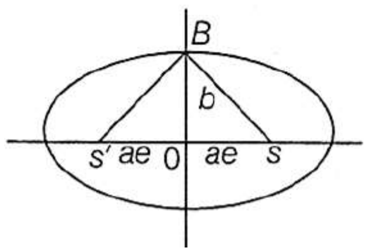
Solving this we get

m $= \frac{-2}{3}$, $\quad m = \frac{-2}{15}$

$\therefore\ y = = \frac{-2}{3}x \pm \sqrt{\frac{3}{2}\times\frac{4}{9}+\frac{1}{3}}$

y$= \frac{-2}{3}x \pm 1$ =====$(3y+2x = \pm 3)$

27. (a) $\frac{x^2}{a^2}+\frac{y^2}{b^2} = 1$



BSS' is an equilateral

$\therefore\ BS' = BS = SS' = 2ae$

Focal property $BS'+BS = 2a$

$BS+BS' = 2SS'$

$2ae+2ae = 2a \Rightarrow 4ae = 2a \Rightarrow e = \frac{1}{2}$

28. (c) $x^2+16y^2=16$

$\frac{x^2}{16}+\frac{y^2}{1}=1$

$a^2=16, b^2=1, m=\tan 60^\circ=\sqrt{3}$

Let tangent be: $y=mx\pm\sqrt{a^2m^2+b^2}$

$y=\sqrt{3}x\pm\sqrt{16\times 3+1}$

$\Rightarrow y=\sqrt{3}x\pm 7$

$\sqrt{3}x-y\pm 7=0$

29. (b) Ellipse $\frac{x^2}{4}+\frac{(y-1)^2}{9}=1$ Circle $x^2+y^2=4$

$9(4-y^2)+4(y-1)^2=36$

$5y^2+8y-4=0$

$y=\frac{2}{5}, y=-2\Rightarrow x=\pm\frac{\sqrt{96}}{5}, x=0$

$\Rightarrow x^2=4-y^2$

Points $\left\{\frac{\sqrt{96}}{5},\frac{2}{5}\right\},\left\{-\frac{\sqrt{96}}{5},\frac{2}{5}\right\},(0,-2)$

∴ 3 Points

30. (c) $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1, a>b$

$\frac{dy}{dx}=-\frac{x}{y}\frac{b^2}{a^2}$

and $x^2-y^2=c^2$

$2x-2y\frac{dy}{dx}=0\Rightarrow\frac{dy}{dx}=\frac{x}{y}$

For orthogonal intersection $m_1\times m_2=-1$

$\frac{dy}{dx}\times\frac{dy}{dx}\Rightarrow-\frac{b^2}{a^2}\frac{x}{y}\frac{x}{y}=-1$

$b^2x^2=a^2y^2\Rightarrow\frac{x^2}{a^2}=\frac{y^2}{b^2}$

Put this value in ellipse, we get

$\frac{2x^2}{a^2}=1\Rightarrow x^2=\frac{a^2}{2}\Rightarrow x=\pm\frac{a}{\sqrt{2}}$

Hence,

$y^2=\frac{b^2}{2}\quad\Rightarrow y=\pm\frac{b}{\sqrt{2}}$

Put these values in hyperbola, we get

$x^2-y^2=c^2$

$\frac{a^2}{2}-\frac{b^2}{2}=c^2$

Hence, $a^2-b^2=2c^2$

OR

If ellipse and hyperbola are orthogonal it means they are confocal it means they have same coordinates of focus.

$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1 a>b\ e=\frac{\sqrt{a^2-b^2}}{a}$

Its foci are $(\pm ae,0)$

And foci of $x^2-y^2=c^2$ are $(\pm\sqrt{2}c,0)$

$\frac{a\sqrt{a^2-b^2}}{a}=\sqrt{2}c\Rightarrow a^2-b^2=2c^2$

31. (c) For focus of parabola $4(y^2-5y)=-12x-67$

$4\left(y^2-2\left(\frac{5}{2}\right)y+\frac{25}{4}\right)=-12x-67+25=-12x-42$

$4\left(y-\frac{5}{2}\right)^2=-12\left(x+\frac{7}{2}\right)$

$\Rightarrow\left(y-\frac{5}{2}\right)^2=-3\left(x+\frac{7}{2}\right)$

$\left(y-\frac{5}{2}\right)^2=-4\left(\frac{3}{4}\right)\left(x+\frac{7}{2}\right)$

Hence, vertex will be $\left(-\frac{7}{2},\frac{5}{2}\right)$ and $a=\frac{3}{4}$

Focus will be

$\left(-\frac{7}{2}-\frac{3}{4},\frac{5}{2}\right)\Rightarrow\left(-\frac{17}{4},\frac{5}{2}\right)$

32. (b) $e=\frac{1}{3}$ directrix $x=9=\frac{a}{e}$

Hence, $a=3$

$b^2=a^2(1-e^2)=9\left(1-\frac{1}{9}\right)=8$

Equation of ellipse $\frac{x^2}{9}+\frac{y^2}{8}=1$

$\Rightarrow 8x^2+9y^2=72$

33. (b) $\frac{x^2}{25}+\frac{y^2}{b^2}=1$ (Ellipse)

; $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ (Hyperbola)

$a=5$, A= $\frac{12}{5}, B=\frac{9}{5}$

$e^2=1-\frac{b^2}{a^2}$ $e^2=1+\frac{B^2}{A^2}$

$e^2==1+\frac{81}{144}\Rightarrow\frac{15}{12}=e$

Focus of ellipse : $(ae,0)$ Focus of hyperbola : $(Ae,0)$

$(5e,0)\ \left(\frac{12}{5}\times\frac{15}{12},0\right)=(3{,}0)$

Given:

$(5e,0)=(3,0)$

$5e=3\Rightarrow e=\frac{3}{5}$

$\frac{9}{25}=1-\frac{b^2}{25}$

$\Rightarrow\frac{b^2}{25}=1-\frac{9}{25}$

Hence, $b^2=16$

34. (b) $(x^2+2x+1)-4(y^2-2y+1)=7+1-4$

$\Rightarrow(x+1)^2-4(y-1)^2=4$

$\Rightarrow\frac{(x+1)^2}{4}-\frac{(y-1)^2}{1}=1$

Hence, centre is $(-1{,}1)$

$\Rightarrow b^2=a^2(e^2-1)$

$\Rightarrow 1==4(e^2-1)$

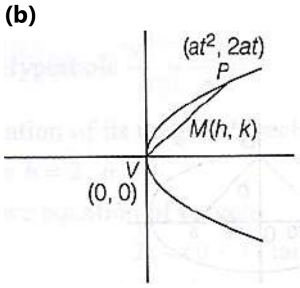
$\Rightarrow$ e= $\frac{\sqrt{5}}{2}$

$\Rightarrow$ ae $=\sqrt{5}$

Foci are at a distance ae from centre.

Hence, foci will be $((-1\pm\sqrt{5}),1)$

35. (b)



Let M be the mid-point of VP is (h,k)

$h=\frac{0+at^2}{2}\Rightarrow at^2=2h$

$k=\frac{0+2at}{2}\Rightarrow at=k\Rightarrow t=\frac{k}{a}$

Put value of t in Eq. (i), we get

$k^2=2ah$

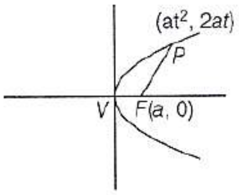
Replace $h\to x$ and $k\to y$

$y^2=2ax$

Put $a=1$

Now, we get $y^2=2x$

36. (b)



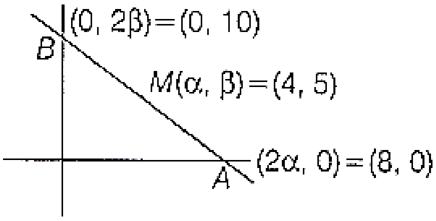
$FP = 5a$
Using focal property
$a + at^2 = 5a$
$I + t^2 = 5$
$\Rightarrow\ t^2 = 4 \Rightarrow t = \pm 2 \Rightarrow t > 0 \Rightarrow t = 2$
Hence, $P \equiv (4a, 4a)$
From given options only $(1,1)$ satisfy P.

37. (c) Let C_1 (h, k) be the center of the circle touches the x axis then into radius is r_1 = k
Circle C_1 touches another circle C_2 (0,2) with radius $r_2 = 2$
$|C_1C_2| \Rightarrow r_1 + r_2$
$\sqrt{(h-o)^2 + (K-3)^2}$ = K + 2
$h^2 - 10K + 5 = 0$, replacing h, k with a, y
Locus is $x^2 - 10y + 5 = 0$
This represents the parabola.

38. **(c)** If it lies in the fourth quadrant, we get $(x, -x)$
Put $y = -x$ in both straight lines.
$2ax + c = 0$ and $3bx + d = 0$
$\Rightarrow x = -\frac{c}{2a}$ and $x = -\frac{d}{3b} \Rightarrow \frac{c}{2a} = \frac{d}{3b} \Rightarrow 3bc - 2ad = 0$

39. (a) Note that area of the parallelogram bounded by the lines $y = m_1x + c_1, y = m_1x + c_2$ and $y = m_2x + d_1$, $y = m_2x + d_2$ is given by $\left|\frac{(c_1-c_2)(d_1-d_2)}{m_1-m_2}\right|$
Area of the parallelogram
$y = 4x + 0$ $\quad$ $y = 4x + 1$
$y = -x + 0$ $\quad$ $y = -x + 1$
$c_1 = 0, c_2 = 1$ $\quad$ $d_1 = 0, d_2 = 1$
$m_1 = 4,$ $\quad$ $m_2 = -1$
$\left|\frac{(c_1-c_2)(d_1-d_2)}{m_1-m_2}\right| = \left|\frac{(0-1)(0-1)}{4-(-1)}\right| = \frac{1}{5}$

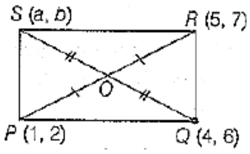
40. (d)
If (α, β) be the mid point of intercepted part AB on axes, then $A(2\alpha, 0), B(0, 2\beta)$ equation of AB will be $\frac{x}{2\alpha} + \frac{y}{2\beta} = 1$. Applying same concept



$\frac{x}{8} + \frac{y}{10} = 1 \Rightarrow 5x + 4y = 40$

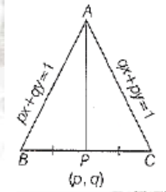
41. (d) $x^2 + y^2 - 2y = 0$ centre $(0,1)$ radius $(r) = 1$
$x^2 + y^2 - 2y - 3 = 0$ centre $(0,1)$ radius $(R) = 2$
Sample space $\pi R^2 = \pi \times 2^2 = 4\pi$
Then, the probability that the point has been taken from smaller circle is
Favourable cases $= \pi r^2 = \pi \times 1^2 = \pi$
Probability $= \frac{\pi}{4\pi} = \frac{1}{4}$

42. **(a)**
In a parallelogram diagonals bisect each other.
O is mid point of PR and SQ.



$O(x, y) = \left(3, \frac{9}{2}\right)$
$\therefore\ \left(3, \frac{9}{2}\right) = \left(\frac{a+4}{2}, \frac{b+6}{2}\right)$
$a = 2, b = 3$

43. **(c)**
$AB = px + qy = 1;$
$AC = qx + py = 1$
$AP: AB + \lambda AC = 0$
i.e. $(px + qy - 1) + \lambda(qx + py - 1) = 0$



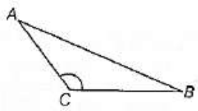
Now Eq. (i) passes through (p, q)
$\Rightarrow (p^2 + q^2 - 1) + \lambda(2pq - 1) = 0$
$\lambda = -\left(\frac{p^2+q^2-1}{2pq-1}\right)$
$\therefore AP: (px + qy - 1) - \left(\frac{p^2+q^2-1}{2pq-1}\right)(qx + py - 1) = 0$
$\Rightarrow (2pq - 1)(px + qy - 1) - (p^2 + q^2 - 1)$
$(qx + py - 1) = 0$

44. (d) Let a and b be the sides of triangle

$a + b = x \Rightarrow ab = y$

Given, $x^2 - z^2 = y$

Here, z is its third side.



$(a + b)^2 - z^2 = ab \Rightarrow a^2 + b^2 + 2ab - z^2 = ab$
$a^2 + b^2 + ab = z^2 \Rightarrow a^2 + b^2 - z^2 = -ab$
$\frac{a^2+b^2-z^2}{2ab} = -\frac{1}{2} \Rightarrow \cos C = \frac{2\pi}{3}$
So, the triangle is obtuse angled.

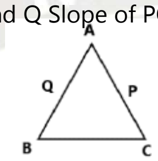
45. **(b)**
$S_1: x^2 + y^2 = 4$, $S_2: x^2 + y^2 - 6x - 8y = 24$
$C_1(0,0),\ C_2(3,4)$
$r_1 = 2\ r_2 = 7$
$\therefore C_1C_2 = |r_1 - r_2| = 5$

$\because$ Circle touches internally.
$\therefore$ Number of common tangents = 1

46. **(c)** $x^2 + y^2 + 2gx + 2fy = 0$
$C_1(-g, -f), r_1 = \sqrt{g^2 + f^2}$
$x^2 + y^2 + 2g'x + 2f'y = 0$
$C_2(-g', -f'), r_2 = \sqrt{g'^2 + f'^2}$
When two circle touches, $r_1 + r_2 = C_1C_2$
$\sqrt{g^2 + f^2} + \sqrt{g'^2 + f'^2} = \sqrt{(g - g')^2 + (f - f')^2}$
Squaring both sides, we get
$g^2 + g'^2 + f^2 + f'^2 + 2\sqrt{g^2 + f^2}\sqrt{g'^2 + f'^2}$
$= g^2 + g'^2 - 2gg' + f^2 + f'^2 - 2ff'$
$(g^2 + f^2)(g'^2 + f'^2) = g^2g'^2 + f^2f'^2 + 2gg'ff'^2$
$g^2f'^2 + f^2g'^2 = 2gg'ff'$
$(gf' - fg')^2 = 0$
$gg' = ff'$

47. (b) Let x coordinate of D is λ
y. l-λ
CD⊥AB $(m_1m_2 = -1)$
Slope of CD $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{1-\lambda}{\lambda - 2}$ $m_1 = -1$
$m_2 = -3-\lambda$
λ-2
$m_1 - m_2 = -1$
$-1 - \left(\frac{3+\lambda}{\lambda-2}\right) = -1$
3 + λ = 2–λ
2λ = –1
$x = \lambda = -\frac{1}{2}$
$y = 1 + \frac{1}{2}$
$y = \frac{3}{2}$ (b) option

48. (b) (2,1), (-1, -2) (3,3) are the mid points to sides BC, CA, AB of ΔABC
Let the mid points of lie CA and AB are P and Q Slope of PQ
$m = \frac{y_0 - y_1}{x_2 - x_1} = \frac{3+2}{3+1} = \frac{5}{4}$



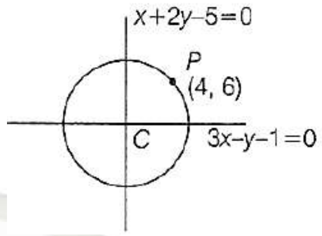
Line posing through (2, 1)
And parallel to line with slope 5/4.
$y - 1 = \frac{5}{4}(x-2)$
5x -4y – 6 = 0

49. (b) The equation of perpendicular line to 7x – y + 8 = 0 is x + 7 y = λ ....(1)
–4 + 7 (5) = λ
λ = 31
Equation will be, x + 7y = 31

50. (a) Let C_1 (h, k) be the center of the circle touches the x axis then into radius is r_1 = k
Circle C_1 touches another circle C_2 (0,2) with radius r_2 = 2
$|C_1C_2| \Rightarrow r_1 + C_2$
$\sqrt{(h - o)^2 + (K - 3)^2} = K + 2$
$h^2 - 10K + 5 = 0$, replacing h, k with a, y
Locus is $x^2 - 10y + 5 = 0$

51. **(d)** Circle $S_1, x^2 + y^2 = 16$
Circle $S_2, x^2 + y^2 - 2y = 0$
Centre C_1 is $C_1(0,0)$ and radius $r_1 = 4$
Centre C_2 is $C_2(0,1)$ and radius $r_2 = 1$
Distance between centres is $C_1C_2 = 1$
Since, $C_1C_2 < |r_2 - r_1|$
Circle S_2 is completely inside S_1
$\therefore$ No common tangents

52. **(c)** Center 3x + 6y – 15 = 0, 3x – y – 1 = 0
y = 2 and x = 1



Centre (1,2)
Radius = CP = $\sqrt{(4-1)^2 + (6-2)^2} = \sqrt{9+16} = 5$
Equation $(x-1)^2 + (y-2)^2 = 25$
$\Rightarrow x^2 + y^2 - 2x - 4y - 20 = 0$

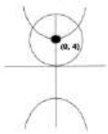
53. (d) Given equations of line are
x + (a –1) y+ 1 = 0 ....(i)
and 2x + a^2y – 1 = 0 ....(ii)
since, line (i) and line (ii) are perpendicular
∴ slope of line 1x
Slope of line 2 = –1
∴ slope of line x + (a–1) y + 1 = 0
$m_1 = \frac{-1}{a-1}$
Slope of line zx + a^2 y - 1 =0
$m_2 = \frac{-2}{a^2}$
$\because$ $m_1m_2 = -1$
$\frac{1}{(a-1)} \frac{2}{a^2} = -1$
= 2 = $-a^2$ (a-1)
= 2 = – a^3 + a^2
= a^3 + a^2 –2 = 0 ....(iii)
On solving Equation (iii) we get
a = –1

54. (c) As we know that the pair of straight line
ax^2 + 2 hxy + by^2 + 2gx +2fy + c = 0 intersect each other at
$\left(\frac{hf-bg}{ab-h^2}, \frac{hg-af}{ab-h^2}\right)$
Therefore, points of intersection of pair of straight line
x^2 + 3xy + $2y^2$ – x – 4y – 6 = 0 is
$\left\{\frac{\frac{3}{2}\times(-2)-(2)\left(-\frac{1}{2}\right)}{(1\times2)-\left(\frac{3}{2}\right)^2}, \frac{\frac{3}{2}\times\left(-\frac{1}{2}\right)-1\times(-2)}{1\times2-\left(\frac{3}{2}\right)^2}\right\}$
$= \frac{-3+1}{2-\frac{9}{4}}, \frac{-\frac{3}{4}+2}{2-\frac{9}{4}}$
$= \left(\frac{-2}{-\frac{1}{4}}, \frac{\frac{5}{4}}{-\frac{1}{4}}\right) = (8, -5)$

55. (b)
y^2 = 40 (x +c) I
y^2 = 4b x II
The eq. of normal far I
y = m (x+c) – (am^3 + 2am)
for eq. 2 normal
y = mx – (bm^3 + 2bm)

If the are common normal then
$mx-(bm^3+2bm)=m(n+c)-(am^3+2an)$
$(a-b)m^3+m(2a-2b)=mc$
$m((a-b)m^2+2a-2b-c)=0$
for common normal
m = 0 or
$m^2=\frac{-(2a-2b-c)}{a-b}$
$m=\sqrt{-2+\frac{c}{a-b}}$
$-2+\frac{c}{a-b}>0$
$\frac{c}{a-b}>2$
$c>2(a-b)$
option (b)

56. (c) $\frac{y^2}{9}-\frac{x^2}{16}=1$ and $\frac{x^2}{4}+\frac{(y-4)^2}{16}=1$
Only 2 point



57. Foci (4, 3) and (1, 2, 5)

Distance between Cofii = 2a e

SS′ = 2ac = $(12.4)^2+(5.3)^2$

$\sqrt{64+4}$ = $\sqrt{125}$

2ae = $5\sqrt{5}$ $\sqrt{68}$

ae = $\sqrt{17}$

OS′ + OS′ = 2a

$\sqrt{16+9}+\sqrt{144+25}$ = 2a

2a = 5 + 13

A= 9

Ac = $\sqrt{17}$

e = $\frac{\sqrt{17}}{a}=\frac{\sqrt{17}}{9}$

58. (b) $3x^2$ + 10 xy + $11y^2$+ + 148

+ 12y + 5 = 0 ......

a....

a = 3, h = 5, b =11, g =

f = 6, C = 5

h^2– ab

25 –33

-8 < 0

h^2– ab < 0, ellipse

59. (a)

60. (a)

- Let center be $(a,a)\rightarrow$ radius = a (since it touches both axes).

- Distance from (a,a) to line $x-y-2=0$ must be a:

$$\frac{|a-a-2|}{\sqrt{1^2+(-1)^2}}=a\rightarrow\frac{2}{\sqrt{2}}=a\rightarrow a=\sqrt{2}$$

- Area = $\pi a^2=\pi(\sqrt{2})^2=2\pi$

Ans. 2π

61. (a)

Let A $(0,0),B(h,k)C(h,-k)$ on $y^2=x\Rightarrow h=k^2$

Equilateral: $\sqrt{h^2+k^2}=2k$

$\Rightarrow\sqrt{k^4+k^2}=2k$

$\Rightarrow k^2=3\Rightarrow h=3.$

Centroid = $\left(\frac{0+3+3}{3},0\right)=(2,0)$

62. (b)

63. (b)
Find intercepts quickly:
$\frac{x}{y}+\frac{y}{2}=1$
A = $(4,0),B=(0,2)$
midpoint:
M = $\left(\frac{4+0}{2},\frac{0+2}{2}\right)=(2,1)$
Reflection across $x+y=1$:
Shortcut for reflection over $x+y=1$; swap slope along perpendicular and shift:
$$M'=(0,-1)$$
P on $x+y=1$ with AP - BP:
Let P = $(t,1-t)$, solve $AP^2-BP^2\rightarrow$ QUICK: $2T^2-10T+17+2T+1$
$t=\frac{4}{3}$
So P = $\left(\frac{4}{3},-\frac{1}{3}\right)$
distance M'P:
d = $\sqrt{(4/3-0)^2+(-1/3+1)^2}$
$=\sqrt{16/9+4/9}$
$=\frac{2\sqrt{5}}{3}$